The consistency of the BIC Markov order estimator

Article *in* The Annals of Statistics · March 1999

DOI: 10.1214/aos/1015957472 · Source: OAI

The consistency of the BIC Markov order estimator.

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Abstract. The Bayesian Information Criterion (BIC) estimates the order of a Markov chain (with finite alphabet A) from observation of a sample path $x_1, x_2, \ldots, x_n$, as that value $k = \hat{k}$ that minimizes the sum of the negative logarithm of the k-th order maximum likelihood and the penalty term $\frac{|A|^k(|A|-1)}{2}\log n$. We show that $\hat{k}$ equals the correct order of the chain, eventually almost surely as $n \to \infty$, thereby strengthening earlier consistency results that assumed an apriori bound on the order. A key tool is a strong ratio-typicality result for Markov sample paths. We also show that the Bayesian estimator or minimum description length estimator, of which the BIC estimator is an approximation, fails to be consistent for the uniformly distributed i.i.d. process.

AMS 1991 subject classification: Primary 62F12, 62M05; Secondary 62F13, 60J10

Key words and phrases: Bayesian Information Criterion, order estimation, ratio-typicality, Markov chains.

[1]Supported in part by a joint NSF-Hungarian Academy grant 92.
[2]Supported in part by a joint NSF-Hungarian Academy grant INT-9515485.

1 Introduction

A Markov chain is a discrete stochastic process $\{X_n\colon n \geq 1\}$ with values in a set A, called the alphabet, of cardinality $|A| < \infty$, for which there is a $k \geq 1$ such that

$$\mathrm{Prob}(X_1^n = x_1^n) = \mathrm{Prob}(X_1^k = x_1^k) \prod_{i=k+1}^{n} Q(x_i|x_{i-k}^{i-1}),\ n \geq k,\ x_1^n \in A^n, \tag{1.1}$$

for suitable transition probabilities $Q(\cdot|\cdot)$. Here and in the sequel, x_m^n denotes the sequence $x_m, x_{m+1}, \ldots, x_n$. The class of processes such that (1.1) holds for a given $k \geq 1$ will be denoted by $\mathcal{M}_k$, and $\mathcal{M}_0$ will denote the class of i.i.d. processes. The *order* of a process in $\mathcal{M} = \cup_{k=0}^{\infty}\mathcal{M}_k$ is the smallest integer k_0 such that for some $\ell \geq 1$, $\{X_n\colon n \geq \ell\}$ is in $\mathcal{M}_{k_0}$.

An important problem is to estimate the order of a Markov chain from observation of a finite sample path. A popular method is the so-called Bayesian Information Criterion (BIC) of [14], which gives the estimator defined by

$$\hat{k}_{\mathrm{BIC}} = \hat{k}_{\mathrm{BIC}}(x_1^n) = \arg\min_k\Big(-\log P_{\mathrm{ML}(k)}(x_1^n) + \frac{|A|^k(|A|-1)}{2}\log n\Big), \tag{1.2}$$

where $P_{\mathrm{ML}(k)}(x_1^n)$ is the k-th order maximum likelihood, i.e., the largest probability given to x_1^n by processes in $\mathcal{M}_k$.

In this paper we address the consistency problem for the BIC estimator, that is, the question of whether $\hat{k}_{\mathrm{BIC}} = k_0$, eventually almost surely, for any process in $\mathcal{M}$. Our principal result is that consistency holds for the subclass of irreducible processes, that is the class of processes that are i.i.d. or belong to $\mathcal{M}_k$ for some $k \geq 1$ and have the additional property that the k-blocks that occur with positive probability communicate.

The BIC consistency theorem.
For any irreducible process in $\mathcal{M}$, $\hat{k}_{\mathrm{BIC}}(X_1^n)$ is eventually almost surely equal to the order of the process.

The restriction to the irreducible case is justifiable for two reasons. First, if transient blocks were permitted, the order of the process could depend on those, and then it could not be estimated consistently. The definition of order could, however, be modified so as to keep the theorem valid for this case also. Second, the theorem is not true if the process is a mixture of two processes of different orders, because the sample paths drawn from the lower order component have positive probability and on these the BIC estimator is eventually almost surely equal to the lower order.

Claims of BIC consistency have been made by several authors, see, for example [5, 2] as well as several papers listed in [2]. All these papers include the explicit or hidden assumption that there are only a finite number of model classes; formally, the minimization in (1.2) is for $k \le k^*$, where k^* is an upper bound on the order of the process. In this case, however, as noted in [8, 5], $\log n$ can be replaced in (1.2) by something much smaller, e. g., a suitable multiple of $\log\log n$. Our theorem does *not* assume a finite number of model classes. The minimization in (1.2) is over all k, though, obviously, it may be restricted to $k < n$.

We should point out that Kieffer established consistency without an apriori bound on the order for the case when $|A|^k(|A|-1)$ is replaced in (1.2) by a more rapidly growing function of k, [9]. Kieffer also raised the question which is answered by our consistency theorem. We also mention the paper of Papangelou [12] whose flavor is somewhat similar to our work, although a direct relationship of results is not apparent.

In the sequel, processes will be identified with their distributions. Thus, if a probability measure P on A^∞ is the distribution of a process in $\mathcal{M}$, we will write $P \in \mathcal{M}$ and say that P is a process in $\mathcal{M}$. Note that to each each irreducible $P \in \mathcal{M}$ there exists a stationary (i. e., shift-invariant) $Q \in \mathcal{M}$ whose order and transition matrix are the same as those of P. This "stationary modification" of the process P is obtained by replacing $\mathrm{Prob}(X_1^k = x_1^k)$ in (1.1) by the limit as $n \to \infty$ of the arithmetic mean of the probabilities $\mathrm{Prob}(X_i^{i+k-1} = x_1^k)$, $i = 1, \ldots, n$. Any irreducible process in $\mathcal{M}$ is absolutely continuous with respect its stationary modification, hence it suffices to prove the consistency theorem for stationary irreducible processes in $\mathcal{M}$.

An essential tool in proving the consistency of the BIC order estimator is a strong ratio-typicality result we establish for stationary, irreducible Markov chains, a result that appears to be of independent interest. It says, loosely, that as long as k does not grow too rapidly with sample path length, the ratio of the empirical relative frequency of each k-block to its probability is uniformly close to 1. The empirical distribution of k-blocks in x_1^n, or the *k-type of x_1^n*, is defined by the formula

$$\hat{P}_n(a_1^k) = \frac{1}{n-k+1} N(a_1^k|x_1^n),\ a_1^k \in A^k, \tag{1.3}$$

where $N(a_1^k|x_1^n) = |\{i \in [1, n-k+1]\colon\ x_i^{i+k-1} = a_1^k\}|$ is the number of occurrences of a_1^k in x_1^n.

The sequence x_1^n is called *(k,ϵ)-typical* for a process Q if $\hat{P}_n(a_1^k) = 0$, whenever $Q(a_1^k) = 0$, and

$$\left|\frac{\hat{P}_n(a_1^k)}{Q(a_1^k)} - 1\right| < \epsilon, \text{ whenever } Q(a_1^k) > 0. \tag{1.4}$$

The typicality theorem.

For any stationary irreducible process $Q \in \mathcal{M}$, there exist positive numbers α and β such that eventually almost surely as $n \to \infty$, the sequence x_1^n is $(k, n^{-\beta})$-typical for every $k \le \alpha \log n$.

Earlier typicality results in which block length grows with sample path length include the following.

1. Marton and Shields obtain large deviations bounds for the variational distance between the empirical k-block distribution and the theoretical k-block distribution that are valid for all $k \le (\log n)(H+\epsilon)$, where H is the process entropy, [11]. Their results do yield our typicality theorem for $k = o(\log\log n)$, but this is not sufficient for our proof of the BIC consistency theorem.

2. Flajolet, Kirschenhofer, and Tichy have obtained similar results for the case when Q is the unbiased coin-tossing process, [7]. They consider longer blocks but do not give an error rate.

The typicality theorem will be used to show that, eventually almost surely, $\hat{k}_{\mathrm{BIC}}(x_1^n)$ cannot belong to the interval $(k^*, \alpha \log n)$, where k^* and $\alpha > 0$ are constants depending on the given process. In addition to the typicality theorem, the proof of this uses a counting argument to bound probabilities, known in information theory as the method of types. For previous applications of the method of types to Markov order estimation see [6].

A second idea, which comes from a more careful look at the Bayesian framework that underlies the BIC, will be used to show that

$$\frac{1}{\log n}\hat{k}_{\mathrm{BIC}}(x_1^n) \to 0, \text{ a. s.} \tag{1.5}$$

This then combines with the result of the preceding paragraph, and with the known consistency results, subject to $k \le k^*$ in (1.2), to establish the BIC consistency theorem.

The Bayesian framework starts with a prior distribution $\{p_k\}$ on the possible orders, together with a prior distribution on $\mathcal{M}_k$, for each k; the latter defines a mixture distribution, that is, a weighted average of the processes in $\mathcal{M}_k$. The Bayesian order estimator is that k for which the product of p_k and the mixture probability of x_1^n is largest. We consider here the mixture of those k-th order Markov chains whose starting distribution is uniform on A^k, taking as prior the $|A|^k$-fold product of the $(\frac{1}{2}, \ldots, \frac{1}{2})$ Dirichlet distributions put on the transition probability matrices $Q(\cdot|\cdot)$. This mixture distribution plays a distinguished role in the theory of universal coding, see Krichevsky and Trofimov, [10]. We denote it by KT_k; its explicit form is given later. The expression minimized in the definition (1.2) of $\hat{k}_{\mathrm{BIC}}(x_1^n)$ is an approximation to $-\log \mathrm{KT}_k(x_1^n)$. We will show, however, that for large n and k, it substantially overestimates $-\log \mathrm{KT}_k(x_1^n)$, a fact that easily leads to the result (1.5).

In contrast to the BIC consistency theorem, for the Bayesian order estimator

$$\hat{k}_{\mathrm{KT}}(x_1^n) = \arg\min_k \Big(-\log p_k - \log \mathrm{KT}_k(x_1^n) \Big), \tag{1.6}$$

we will prove the following.

The inconsistency theorem.

If $\{X_n\}$ is i.i.d. with X_i uniformly distributed on A, then $\hat{k}_{\mathrm{KT}}(x_1^n) \to +\infty$, almost surely, provided the p_k of (1.6) are taken, as usual, to be slowly decreasing, say $p_k = ck^{-2}$.

Similar phenomena have been established in other Bayesian settings by Diaconis and Freedman, see, for example, [4]. We note that the order estimator $\hat{k}_{\mathrm{KT}}(x_1^n)$ is often introduced via the minimum description length (MDL) principle, see Rissanen, [13], or Barron, Rissanen, and Yu, [2]. It is interesting to compare our inconsistency theorem with an MDL-inspired result by Barron, [1], see also [2]. That result, specialized to the estimator $\hat{k}_{\mathrm{KT}}(x_1^n)$, can be formulated as follows. If the processes of $\mathcal{M}_k$ are parametrized by a $|A|^k(|A|-1)$ dimensional parameter, then for Lebesgue almost all choices of the parameter, the estimator $\hat{k}_{\mathrm{KT}}(x_1^n)$ is eventually almost surely equal to the correct order. Our inconsistency theorem shows that "for almost every choice of the parameter," is not vacuous.

We note that the MDL principle also leads to non-Bayesian estimators, replacing, e. g., mixture distributions by normalized maximum likelihoods, see [2]. In our context this means replacing $\mathrm{KT}_k(x_1^n)$ by

$$P_{\mathrm{ML}(k)}(x_1^n) \Big/ \sum_{y_1^n \in A^n} P_{\mathrm{ML}(k)}(y_1^n).$$

We show the inconsistency theorem also holds for the resulting order estimator.

2 Proof of the BIC consistency theorem.

The notation $a_1^k \in x_1^n$ will mean that $N(a_1^k|x_1^n) > 0$, where, as defined earlier, $N(a_1^k|x_1^n)$ is the number of occurrences of a_1^k in x_1^n. Equation (1.1) can then be expressed as

$$\mathrm{Prob}(X_1^n = x_1^n) = \mathrm{Prob}(X_1^k = x_1^k) \prod_{a_1^{k+1} \in x_1^n} Q(a_{k+1}|a_1^k)^{N(a_1^{k+1}|x_1^n)}. \tag{2.1}$$

For fixed $x_1^n \in A^n$, this probability is maximized if $\mathrm{Prob}(X_1^k = x_1^k) = 1$ and $Q(a_{k+1}|a_1^k) = N(a_1^{k+1}|x_1^n)/N(a_1^k|x_1^{n-1})$, whenever $a_1^k \in x_1^{n-1}$. Thus the logarithm of the k-th order maximum likelihood is given by the formula

$$\log P_{\mathrm{ML}(k)}(x_1^n) = \sum_{a_1^{k+1} \in x_1^n} N(a_1^{k+1}|x_1^n) \log \frac{N(a_1^{k+1}|x_1^n)}{N(a_1^k|x_1^{n-1})} \tag{2.2}$$

where $N(a_1^k|x_1^{n-1})$ is replaced by n in the case when $k = 0$.

As noted in the introduction, if suffices to prove consistency of the BIC estimator for stationary irreducible processes in $\mathcal{M}$. This will be done by establishing two propositions. In the statement of these and in the remainder of this section Q will denote a stationary irreducible process in $\mathcal{M}$ of order k_0 and all almost sure statements are with respect to Q.

Proposition 1

There exist positive numbers k^* and α, both depending on Q, such that $\hat{k}_{\mathrm{BIC}}(x_1^n) \notin (k^*, \alpha \log n)$, eventually almost surely.

Proposition 2

$\frac{1}{\log n}\hat{k}_{\mathrm{BIC}}(x_1^n) \to 0$, *almost surely.*

These two results, combined with the known result, see [5], that

$$\hat{k}_{\mathrm{BIC}}(x_1^n) \notin [0, k_0) \cup (k_0, k^*], \text{ eventually almost surely,} \tag{2.3}$$

imply that $\hat{k}_{\mathrm{BIC}}(x_1^n) = k_0$, eventually almost surely, as desired. For completeness a short proof of (2.3) is given in the Appendix.

2.1 Proof of Proposition 1.

The proof of Proposition 1 uses counting arguments we now develop, together with the typicality theorem, whose proof is delayed to the next section. Recall from (1.3) that the empirical k-block distribution, or k-type, of x_1^n is the distribution on A^k defined by the formula

$$\hat{P}_n(a_1^k) = \frac{1}{n-k+1} N_n(a_1^k | x_1^n).$$

Note that since

$$\sum_{a_{k+1} \in A} N(a_1^{k+1} | x_1^n) = N(a_1^k | x_1^{n-1}),\ a_1^k \in A^k,$$

the k-type of x_1^{n-1} is the marginal of the $(k+1)$-type of x_1^n.

A distribution P on A^k is called a *k-type with path length n* if it equals the k-type of some $x_1^n \in A^n$. The *type class* of P, denoted by $\mathcal{T}_P^n$, is the set of x_1^n that have k-type P. The bounds in Lemma 1 and Lemma 2, below, on the size of typical type classes and the number of typical type classes, respectively, will play a key role in the proof of Proposition 1.

For each j, let $\mathcal{S}_j$ denote the support of the j-marginal of Q, that is , the set of all a_1^j for which $Q(a_1^j) > 0$. A k-type class $\mathcal{T}_P^n$ is ϵ-typical (for the given Q) if some, and hence all, $x_1^n \in \mathcal{T}_P^n$ are (k, ϵ)-typical in the sense of (1.4), i. e., if

$$\left| \frac{P(a_1^k)}{Q(a_1^k)} - 1 \right| < \epsilon,\ a_1^k \in \mathcal{S}_k, \quad P(a_1^k) = 0,\ a_1^k \in A^k \setminus \mathcal{S}_k.$$

For convenience, we state our key lemma in terms of $(k+1)$-type classes rather than k. In its statement $P(\cdot | a_1^k)$ denotes the distribution on A defined by $P(a_{k+1} | a_1^k) = P(a_1^{k+1})/P(a_1^k)$, and $H(P(\cdot | a_1^k))$ denotes the Shannon entropy (with natural logarithms) of $P(\cdot | a_1^k)$.

Lemma 1

There is a positive constant $C = C(Q)$ such that

$$|\mathcal{T}_P^n| \leq C^{|A|^k}(n-k)^{-\frac{|\mathcal{S}_{k+1}|-|\mathcal{S}_k|}{2}} \exp\Big(-(n-k)\sum_{a_1^k\in\mathcal{S}_k} P(a_1^k)H(P(\cdot|a_1^k))\Big),$$

for every ϵ-typical $(k+1)$-type class $\mathcal{T}_P^n$, for all $\epsilon < 1/2$, for all n, and for all $1 \leq k < n$.

Proof. The proof uses a bound on the size of 1-type classes that follows from the refined version of Stirling's formula, see [3, Exercise 2, §1.2]. Let P be a 1-type with path length m and support $\mathcal{S}(P) = \{a \in A : P(a) > 0\}$. The bound states that

$$|\mathcal{T}_P^m| \leq \frac{(2\pi m)^{-\frac{|\mathcal{S}(P)|-1}{2}}}{\prod_{a\in\mathcal{S}(P)}\sqrt{P(a)}} \exp\Big(-mH(P)\Big). \tag{2.4}$$

Fix $0 < \epsilon < 1/2$ and $1 \leq k < n$, and let P be an ϵ-typical $(k+1)$-type of path length n. Typicality implies that $P(a_1^k) > 0$ iff $a_1^k \in \mathcal{S}_k$ and $P(a_1^{k+1}) > 0$ iff $a_1^{k+1} \in \mathcal{S}_{k+1}$. Furthermore, denoting the support of the conditional distribution $P(\cdot|a_1^k)$ by $\mathcal{S}(\cdot|a_1^k)$, we have

$$\sum_{a_1^k\in\mathcal{S}_k} \frac{|\mathcal{S}(\cdot|a_1^k)|-1}{2} = \frac{|\mathcal{S}_{k+1}|-|\mathcal{S}_k|}{2}. \tag{2.5}$$

Given $x_1^n \in \mathcal{T}_P^n$, let $x^*(a_1^k)$ denote the sequence of length $(n-k)P(a_1^k)$ consisting of the symbols x_{t_j} that follow the occurrences of a_1^k in x_1^{n-1}, i. e., such that $x_{t_j-k}^{t_j-1} = a_1^k$, $k < t_j \leq n$. Then $x^*(a_1^k)$ has 1-type $P(\cdot|a_1^k)$ and it follows that assigning to x_1^n the $|\mathcal{S}_k|$-tuple $\{x^*(a_1^k) : a_1^k \in \mathcal{S}_k\}$ defines a mapping of $\mathcal{T}_P^n$ into the Cartesian product of the sets $\mathcal{T}_{P(\cdot|a_1^k)}^{(n-k)P(a_1^k)}$, $a_1^k \in \mathcal{S}_k$. Furthermore, this mapping is one-to-one when restricted to the set $\mathcal{T}_{P,x_1^k}^n$ consisting of all $x_1^n \in \mathcal{T}_P^n$ that start with a fixed x_1^k. Hence it follows from the 1-type bound (2.4) and the support relations (2.5) that

$$\begin{aligned}
|\mathcal{T}_{P,x_1^k}^n| &\leq \prod_{a_1^k\in\mathcal{S}_k}\Big|\mathcal{T}_{P(\cdot|a_1^k)}^{(n-k)P(a_1^k)}\Big| \\
&\leq \Pi(P)(n-k)^{-\frac{|\mathcal{S}_{k+1}|-|\mathcal{S}_k|}{2}} \exp\Big\{-(n-k)\sum_{a_1^k\in\mathcal{S}_k} P(a_1^k)H(P(\cdot|a_1^k))\Big\},
\end{aligned}$$

where

$$\Pi(P) = \prod_{a_1^k\in\mathcal{S}_k} \frac{(2\pi P(a_1^k))^{-\frac{|\mathcal{S}(\cdot|a_1^k)|-1}{2}}}{\prod_{a_{k+1}\in\mathcal{S}(\cdot|a_1^k)}\sqrt{P(a_{k+1}|a_1^k)}}. \tag{2.6}$$

Typicality and the assumption that $0 < \epsilon < 1/2$ imply that

$$\Pi(P) \leq c^{|A^k|}\Pi(Q), \tag{2.7}$$

where c is a constant, depending only on Q, and $\Pi(Q)$ is defined by replacing P by Q in (2.6). (One can show that $c = \sqrt{6}$ suffices.) The Markovian property, however, implies the existence of $c_* > 0$ such that $Q(a_1^j) \geq c_*^j$, for any j and any $a_1^j \in \mathcal{S}_j$. This fact, together with (2.7), easily yields the desired result. This completes the proof of Lemma 1. $\square$

Lemma 2

The number of ϵ-typical $(k+1)$-type classes $\mathcal{T}_P^n$ is less than
$|A|^{2k}\left(1+2\epsilon(n-k)\right)^{|\mathcal{S}_{k+1}|-|\mathcal{S}_k|}$.

Proof. It suffices to show that, fixing x_1^k and x_{n-k+1}^n, the number of ϵ-typical $(k+1)$-type classes that could contain x_1^n is less than $(1+2\epsilon(n-k))^{|\mathcal{S}_{k+1}|-|\mathcal{S}_k|}$. Note that for a $(k+1,\epsilon)$-typical x_1^n, among the numbers $N(a_1^{k+1}|x_1^n)$ exactly those with $a_1^{k+1} \in \mathcal{S}_{k+1}$ are positive, and each have less than $1+2\epsilon(n-k)$ possible values, because typicality implies

$$\left|N\left(a_1^{k+1}|x_1^n\right)-(n-k)Q(a_1^{k+1})\right| < \epsilon(n-k)Q(a_1^{k+1}) \leq \epsilon(n-k).$$

Hence the proof will be complete if we show that, fixing x_1^k and x_{n-k+1}^n, the numbers $N(a_1^{k+1}|x_1^n)$, $a_1^{k+1} \in \mathcal{S}_{k+1}$, satisfy $|\mathcal{S}_k|$ independent linear constraints, so that $|\mathcal{S}_{k+1}|-|\mathcal{S}_k|$ of these numbers uniquely determine the others.

Clearly, the following equations always hold:

$$\sum_{a_1^{k+1}\in\mathcal{S}_{k+1}} N\left(a_1^{k+1}|x_1^n\right) = n-k,$$

and for each $b_1^k \in \mathcal{S}_k$,

$$\sum_{a_1^{k+1}\in\mathcal{S}_{k+1}} N\left(a_1^{k+1}|x_1^n\right)\left(\mathbb{I}\left(a_1^k = b_1^k\right)-\mathbb{I}\left(a_2^{k+1} = b_1^k\right)\right) = \mathbb{I}\left(a_1^k = x_1^k\right)-\mathbb{I}\left(a_1^k = x_{n-k+1}^n\right),$$

where here and in the remainder of the paper, $\mathbb{I}(\cdot)$ denotes the indicator function.

It is easy to see that dropping any one of the last $|\mathcal{S}_k|$ equations, the remaining $|\mathcal{S}_k|$ constraints are independent. Formally, $\mathbb{I}$ (with each component equal to 1) and $\left\{\mathbb{I}(a_1^k = b_1^k)-\mathbb{I}(a_2^{k+1} = b_1^k), a_1^{k+1} \in \mathcal{S}_{k+1}\right\}$ with b_1^k running over all k-blocks in $\mathcal{S}_k$ but one, are linearly independent vectors in $\mathbb{R}^{|\mathcal{S}_{k+1}|}$. $\square$

Proof of Proposition 1 continued. Put

$$B_{n,k} = \{x_1^n\colon\ \hat{k}_{\mathrm{BIC}}(x_1^n) = k\}.$$

Since Q is Markov of order k_0, we have $Q(x_1^n) \leq P_{\mathrm{ML}(k_0)}(x_1^n)$, so that for $x_1^n \in B_{n,k}$, the definition of $\hat{k}_{\mathrm{BIC}}(x_1^n)$ implies that

$$-\log P_{\mathrm{ML}(k)}(x_1^n)+\frac{|A|^k(|A|-1)}{2}\log n \leq -\log Q(x_1^n)+\frac{|A|^{k_0}(|A|-1)}{2}\log n,$$

that is,

$$\log Q(x_1^n) \le \log P_{\mathrm{ML}(k)}(x_1^n) - \frac{|A|^k(|A|-1)}{2}\log n + \frac{|A|^{k_0}(|A|-1)}{2}\log n. \qquad (2.8)$$

Let α and β be the positive numbers given by the typicality theorem for Q and let $C = C(Q)$ be the number given by Lemma 1. For any $(k+1)$-type P with path length n the maximum likelihood formula (2.2) and the definition of type (1.3), combine to yield

$$(n-k)\sum_{a_1^k\in\mathcal{S}_k} P(a_1^k)H(P(\cdot|a_1^k)) = -\log P_{\mathrm{ML}(k)}(x_1^n),\ x_1^n \in \mathcal{T}_P^n. \qquad (2.9)$$

This and (2.8), together with Lemma 1, imply that

$$Q(\mathcal{T}_P^n \cap B_{n.k}) \le C^{|A|^k}(n-k)^{-\frac{\Delta_k S}{2}}\exp\Big(-\frac{(|A|^k-|A|^{k_0})(|A|-1)}{2}\log n\Big), \qquad (2.10)$$

for any ϵ-typical $(k+1)$-type class $\mathcal{T}_P^n$ with $\epsilon < 1/2$, where we used the notation $\Delta_k\mathcal{S} = |\mathcal{S}_{k+1}| - |\mathcal{S}_k|$.

Let $G_{n.k}$ be the union of the $n^{-\beta}$-typical $(k+1)$-type classes $\mathcal{T}_P^n$. The bound (2.10) and Lemma 2 combine to show that $Q(G_{n.k}\cap B_{n.k})$ is upper bounded by

$$C^{|A|^k}(1+2n^{-\beta}(n-k))^{\Delta_k\mathcal{S}}(n-k)^{-\frac{\Delta_k S}{2}}\exp\Big(-\frac{(|A|^k-|A|^{k_0})(|A|-1)}{2}\log n\Big),$$

which, using the notation $\Delta_k A = |A|^{k+1} - |A|^k$, is, in turn, upper bounded by

$$(\mathrm{constant})^{|A|^k} n^{-\beta\Delta_k\mathcal{S}}(n-k)^{\frac{\Delta_k S}{2}} n^{-\frac{\Delta_k A}{2}}\exp\Big(\frac{|A|^{k_0}(|A|-1)}{2}\log n\Big). \qquad (2.11)$$

The right-most factor is a polynomial in n and

$$n^{-\beta\Delta_k\mathcal{S}}(n-k)^{\frac{\Delta_k S}{2}} n^{-\frac{\Delta_k A}{2}} \le n^{-(\Delta_k A - \Delta_k S(1-2\beta))/2} \le n^{-\beta\Delta_k A},$$

since $0 \le \Delta_k S \le \Delta_k A$, and we can assume that $\beta < 1/2$, because once the typicality theorem holds for a given β it holds for all smaller β. If k is sufficiently large, the bound $n^{-\beta\Delta_k A}$ dominates the bound (2.11), since $\beta > 0$, and $\Delta_k A = |A|^k(|A|-1)$ goes to infinity. Hence, there is a $k^* \ge k_0$ and an n^* such that

$$Q(G_{n.k}\cap B_{n.k}) \le n^{-2},\ n \ge n^*,\ k^* \le k \le \alpha\log n,$$

from which it easily follows that

$$\sum_n Q\Big(\bigcup_{k\in[k^*,\alpha\log n]} G_{n.k}\cap B_{n.k}\Big) < \infty.$$

The typicality theorem implies

$$x_1^n \in \bigcup_{k\in[k^*,\alpha\log n]} G_{n.k},\ \text{eventually a.s.},$$

and an application of Borel-Cantelli yields Proposition 1. $\square$

2.2 Proof of Proposition 2.

Again we use the notation $a_1^k \in x_1^n$ to mean that $N(a_1^k|x_1^n) > 0$. The proof of Proposition 2 relies on properties of the Krichevsky-Trofimov distributions KT_k. For $k = 0$, KT_0 is the $(\frac{1}{2}, \ldots, \frac{1}{2})$-Dirichlet mixture of the i. i. d. distributions with alphabet A. In explicit terms, see [10],

$$\begin{aligned} \mathrm{KT}_0(x_1^n) &= \frac{\Gamma(\frac{|A|}{2})}{\Gamma(n + \frac{|A|}{2})} \prod_{a \in x_1^n} \frac{\Gamma(N(a|x_1^n) + \frac{1}{2})}{\Gamma(\frac{1}{2})} \\ &= \frac{\prod_{a \in x_1^n} (N(a|x_1^n) - \frac{1}{2})(N(a|x_1^n) - \frac{3}{2}) \ldots (\frac{1}{2})}{(n - 1 + \frac{|A|}{2})(n - 2 + \frac{|A|}{2}) \ldots (\frac{|A|}{2})}. \end{aligned} \tag{2.12}$$

For $k \geq 1$, KT_k is the mixture of the k-th order Markov chains whose starting distribution is uniform on A^k, the mixture taken with the $|A|^k$-fold product of $(\frac{1}{2}, \ldots, \frac{1}{2})$-Dirichlet distributions put on the transition matrices $Q(\cdot|\cdot)$. The explicit form is

$$\mathrm{KT}_k(x_1^n) = \frac{1}{|A|^k} \prod_{a_1^k \in x_1^{n-1}} \left[\frac{\prod_{a_{k+1} : a_1^{k+1} \in x_1^n} (N(a_1^{k+1}|x_1^n) - \frac{1}{2})(N(a_1^{k+1}|x_1^n) - \frac{3}{2}) \ldots (\frac{1}{2})}{(N(a_1^k|x_1^{n-1}) - 1 + \frac{|A|}{2})(N(a_1^k|x_1^{n-1}) - 2 + \frac{|A|}{2}) \ldots (\frac{|A|}{2})} \right]. \tag{2.13}$$

For our purposes (2.12) and (2.13) are taken as the definition of $\mathrm{KT}_k(x_1^n)$.

The next lemma summarizes the properties we shall need. A more precise result than inequality (2.14) appears in [17], for the case $|A| = 2$.

Lemma 3

There is a constant C depending only on the alphabet size $|A|$ such that for every $n \geq 1$ and $x_1^n \in A^n$,

$$\left| \log KT_0(x_1^n) - \sum_{a \in A} N(a|x_1^n) \log \frac{N(a|x_1^n)}{n} + \frac{|A| - 1}{2} \log n \right| \leq C, \tag{2.14}$$

and, for every $k \geq 1$,

$$\left| \log KT_k(x_1^n) - \log P_{\mathrm{ML}(k)}(x_1^n) + \frac{|A| - 1}{2} \sum_{a_1^k \in x_1^{n-1}} \log N(a_1^k|x_1^{n-1}) \right| \leq C|A|^k, \tag{2.15}$$

and

$$\log KT_k(x_1^n) \geq \log P_{\mathrm{ML}(k)}(x_1^n) - \frac{|A|^k(|A| - 1)}{2} \left(1 + \log^+ \frac{n}{|A|^k} \right) - C|A|^k, \tag{2.16}$$

where $\log^+ t = \max(\log t, 0)$.

Proof. The well-known bound (2.14), see [10], easily follows from Stirling's formula for Γ-functions, and (2.15) is an immediate consequence of (2.14). Indeed, the square-bracketed factors in the definition (2.13) of KT_k are of the form of the definition (2.12) of KT_0, with $N(a_1^k|x_1^{n-1})$ and $N(a_1^{k+1}|x_1^n)$ in the role of n and $N(a|x_1^n)$, respectively. Applying (2.14) to each of these factors and recalling formula (2.2), that is,

$$\sum_{a_1^k \in x_1^{n-1}} \sum_{\substack{a_{k+1}: \\ a_1^{k+1} \in x_1^n}} N(a_1^{k+1}|x_1^n) \log \frac{N(a_1^{k+1}|x_1^n)}{N(a_1^k|x_1^{n-1})} = \log P_{\mathrm{ML}(k)}(x_1^n),$$

we obtain (2.15), with a somewhat larger C to take care of the logarithm of the $1/|A|^k$ factor that appears in the definition (2.13) of KT_k.

The final bound follows from (2.15), because, by the concavity of

$$f(t) = \begin{cases} \log t & \text{if } t > 1 \\ t - 1 & \text{if } 0 \le t \le 1, \end{cases}$$

we have

$$\begin{aligned} \sum_{a_1^k \in x_1^{n-1}} \log N(a_1^k|x_1^{n-1}) &\le \sum_{a_1^k \in A^k} f(N(a_1^k|x_1^{n-1})) + |A|^k \\ &\le |A|^k f\left(\frac{n}{|A|^k}\right) + |A|^k \le |A|^k \log^+\left(\frac{n}{|A|^k}\right) + |A|^k. \end{aligned}$$

This completes the proof of Lemma 3. $\square$

To establish Proposition 2 we will prove the somewhat stronger assertion that for $k_n = \frac{\log\log n}{\log|A|}$,

$$\hat{k}_{\mathrm{BIC}}(x_1^n) \notin [k_n, n], \text{ eventually a.s.} \tag{2.17}$$

Towards this end, recall equation (2.8), that is, for $x_1^n \in B_{n,k}$, we have

$$\log Q(x_1^n) \le \log P_{\mathrm{ML}(k)}(x_1^n) - \frac{|A|^k(|A|-1)}{2}\log n + \frac{|A|^{k_0}|A|-1)}{2}\log n.$$

Adding $-\log \mathrm{KT}_k(x_1^n)$ to both sides and applying the bound (2.16) of Lemma 3, we obtain

$$-\log \mathrm{KT}_k(x_1^n) + \log Q(x_1^n) \le -3\log n,\ k \ge k_n,\ n \ge n_0,\ x_1^n \in B_{n,k}.$$

This implies that

$$Q(\cup_{k=k_n}^n B_{n,k}) \le n^{-3} \sum_{k=k_n}^{n} \mathrm{KT}_k(B_{n,k}) \le n^{-2},\ n \ge n_0,$$

which, in turn, implies

$$\sum_{n=n_0}^{\infty} Q(\{x_1^n \colon \hat{k}_{\mathrm{BIC}}(x_1^n) \in [k_n, n]\}) \le \sum_{n=n_0}^{\infty} n^{-2} < \infty.$$

An application of the Borel-Cantelli theorem yields the desired result (2.17). This completes the proof of the BIC consistency theorem. $\square$

3 Proof of the typicality theorem.

In this section Q again denotes the distribution of a stationary irreducible Markov chain $\{X_n\colon\ n \geq 1\}$ of order k_0, and, for each j, $\mathcal{S}_j$ denotes the support of the j-th order marginal of Q, that is , the set of all a_1^j for which $Q(a_1^j) > 0$.

To prove the typicality theorem, it suffices to establish the following two propositions.

Proposition 3

For any fixed k there is a constant C such that

$$\left|\frac{\hat{P}_n(a_1^k)}{Q(a_1^k)} - 1\right| < C\sqrt{\frac{\log\log n}{n}},\ \ a_1^k \in \mathcal{S}_k,\ \ \textit{eventually a.s.}$$

Proposition 4

If α and β are small enough positive numbers then, eventually almost surely,

$$\left|\frac{\hat{P}_n(a_1^k)}{Q(a_1^k)} - \frac{\hat{P}_{n-(k-k_0)}(a_1^{k_0})}{Q(a_1^{k_0})}\right| < n^{-\beta},\ \ a_1^k \in \mathcal{S}_k,\ \ k_0 < k \leq \alpha\log n. \tag{3.1}$$

Indeed the claim of the theorem follows for $k_0 < k \leq \alpha \log n$ from Proposition 3 and Proposition 4 with a slightly smaller β.

Proof of Proposition 3. For each $k \geq k_0$, the law of the iterated logarithm applied to the recurrence times of the (delayed) renewal process $\{\mathbb{I}(X_i^{i+k-1} = a_1^k)\colon\ i \geq 1\}$ shows that the proposition is true, see [5]. The assertion for smaller k obviously follows. $\square$

Proof of Proposition 4. The simplest idea would be to overbound the probability that (3.1) does not hold by some γ_n for which $\sum \gamma_n < \infty$, but this does not appear to be feasible. The same idea, however, works when merging "bad events" between consecutive powers of 2. We proceed by noticing first that whenever (3.1) is violated for some $n \in (2^{i-1}, 2^i]$, there exists $k \in (k_0, \alpha \log 2^i]$ and $a_1^k \in \mathcal{S}_k$ such that

$$\max_{2^{i-1}<n\leq 2^i}\left|\frac{\hat{P}_n(a_1^k)}{Q(a_1^k)} - \frac{\hat{P}_{n-(k-k_0)}(a_1^{k_0})}{Q(a_1^{k_0})}\right| > 2^{-i\beta}, \tag{3.2}$$

Thus, letting $B_i(\alpha,\beta)$ be the event that (3.2) occurs for some $k \in (k_0, \alpha\log 2^i]$ and $a_1^k \in \mathcal{S}_k$, it is enough to show that

$$\sum_{i=1}^{\infty} \mathrm{Prob}(B_i(\alpha,\beta)) < \infty, \tag{3.3}$$

for suitable $\alpha > 0$ and $\beta > 0$.

For brevity in the remainder of the proof we denote the k-block frequencies $N(a_1^k|x_1^n)$ by $N_n(a_1^k)$. Then, using that $\hat{P}_n(a_1^k) = N_n(a_1^k)/(n-k+1)$, where

$$N_n(a_1^k) = \sum_{i=1}^{n-k+1} \mathbb{I}(X_i^{i+k-1} = a_1^k),$$

together with the factorization

$$Q(a_1^k) = Q(a_1^{k_0}) \prod_{j=k_0+1}^{k} Q(a_j|a_{j-k_0}^{j-1}),$$

the inequality (3.2) can be rewritten as

$$\max_{2^{i-1}<n\leq 2^i} \left| N_n(a_1^k) - N_{n-(k-k_0)}(a_1^{k_0}) \prod_{j=k_0+1}^{k} Q(a_j|a_{j-k_0}^{j-1}) \right| > 2^{-i\beta} 2^{i-2} Q(a_1^k). \tag{3.4}$$

The difference $N_n(a_1^k) - N_{n-(k-k_0)}(a_1^{k_0}) \prod_{j=k_0+1}^{k} Q(a_j|a_{j-k_0}^{j-1})$ can be decomposed as the sum

$$\sum_{\ell=1}^{k-k_0} \left[N_{n-\ell+1}(a_1^{k-\ell+1}) - N_{n-\ell}(a_1^{k-\ell}) Q(a_{k-\ell+1}|a_{k-\ell+1-k_0}^{k-\ell}) \right] \prod_{j=k-\ell+2}^{k} Q(a_j|a_{j-k_0}^{j-1}), \tag{3.5}$$

hence, if its absolute value exceeds a positive constant λ, then at least one term of the sum (3.5) has absolute value $> \lambda/(k-k_0)$. Using this fact, the probability of the event (3.4) is upper bounded by the sum, for $1 \leq \ell \leq k-k_0$, of the probabilities of the events

$$\max_{2^{i-1}<n\leq 2^i} \left| N_{n-\ell+1}(a_1^{k-\ell+1}) - N_{n-\ell}(a_1^{k-\ell}) Q(a_{k-\ell+1}|a_{k-\ell+1-k_0}^{k-\ell}) \right| > \frac{2^{-i\beta} 2^{i-2} Q(a_1^{k-\ell+1})}{k-k_0}. \tag{3.6}$$

The key to the remainder of the proof is the observation that, for any $m \in (k_0, k]$, the sequence

$$\{N_n(a_1^m) - N_{n-1}(a_1^{m-1}) Q(a_m|a_{m-k_0}^{m-1}) \colon n \geq m\}$$

is a martingale. Indeed,

$$N_n(a_1^m) - N_{n-1}(a_1^{m-1}) Q(a_m|a_{m-k_0}^{m-1}) = \sum_{j=m}^{n} \Delta_j,$$

where

$$\Delta_j = \mathbb{I}(X_{j-m+1}^j = a_1^m) - \mathbb{I}(X_{j-m+1}^{j-1} = a_1^{m-1}) Q(a_m|a_{m-k_0}^{m-1}),$$

which satisfies $E(\Delta_j|X_1^{j-1}) = 0$. Furthermore,

$$E(\Delta_j^2) = Q(a_1^m)\left(1 - Q(a_m|a_{m-k_o}^{m-1})\right) \leq Q(a_1^m),$$

so Kolmogorov's inequality for martingales gives that

$$\mathrm{Prob}\left\{\max_{n\le 2^i}\left|N_n(a_1^m)-N_{n-1}(a_1^{m-1})Q(a_m|a_{m-k_0}^{m-1})\right|>\lambda\right\}<\frac{2^iQ(a_1^m)}{\lambda^2}.$$

Applying this with $m=k-\ell+1$ and $\lambda=2^{k-i\beta-2}Q(a_1^{k-\ell+1})/(k-k_0)$, it follows that the sum, for $1\le\ell\le k-k_0$, of the probabilities of the events (3.6) is upper bounded by

$$\sum_{\ell=1}^{k-k_0}\frac{2^iQ(a_1^{k-\ell+1})(k-k_0)^2}{[2^{i-i\beta-2}Q(a_1^{k-\ell+1})]^2}\le\frac{2^4k^3}{2^{(1-2\beta)i}Q(a_1^k)}. \tag{3.7}$$

Next let γ be the minimum of $Q(a_1^{k_0})$, over all $a_1^{k_0}\in\mathcal{S}_{k_0}$, and let δ be the minimum of $Q(a_{k_0+1}|a_1^{k_0})$, over all $a_1^{k_0+1}\in\mathcal{S}_{k_0+1}$. For all k and all $a_1^k\in\mathcal{S}_k$, we then have $Q(a_1^k)\ge\gamma\delta^k$, and hence our argument shows that the probability that (3.2) holds for some fixed $k\in(k_0,\alpha\log 2^i]$ and $a_1^k\in\mathcal{S}_k$, is bounded above by $2^4k^3/2^{(1-2\beta)i}\gamma\delta^k$. Summing over $k\in(k_0,\alpha\log 2^i]$ and $a_1^k\in\mathcal{S}_k$ produces

$$\begin{aligned}\mathrm{Prob}(B_i(\alpha,\beta)) &\le \sum_{k_0<k\le\alpha\log 2^i}\frac{2^4k^3|A|^k}{2^{(1-2\beta)i}\gamma\delta^k}\\ &< \frac{2^4(\alpha\log 2^i)^4(|A|/\delta)^{\alpha\log 2^i}}{\gamma 2^{(1-2\beta)i}}=\frac{(2\alpha\log 2)^4}{\gamma}i^4 2^{(2\beta-1+\alpha\log\frac{|A|}{\delta})i}.\end{aligned}$$

It follows from the final bound that if $\alpha>0$ and $\beta>0$ are chosen to satisfy $2\beta+\alpha\log\frac{|A|}{\delta}<1$, then the sum of the probabilities of the events $B_i(\alpha,\beta)$ is indeed finite, that is, the desired result (3.3) holds. This completes the proof of the typicality theorem. $\square$

4 Proof of the inconsistency theorem.

In this section, Q denotes the uniform i.i.d. process,

$$Q(x_1^n)=|A|^{-n},\qquad x_1^n\in A^n.$$

We prove that the Bayesian Markov order estimator with Dirichlet priors, viz.

$$\hat{k}_{\mathrm{KT}}(x_1^n)=\arg\min_k(-\log p_k-\log\mathrm{KT}_k(x_1^n))$$

fails to put out 0, the order of Q, eventually almost surely, provided that $\{p_k\}$ is slowly decreasing, that is, $\log p_k=o(k)$. We prove that, in fact,

$$\hat{k}_{\mathrm{KT}}(x_1^n)\to\infty,\ \ \text{almost surely.} \tag{4.1}$$

The same inconsistency will be established also for the non-Bayesian MDL estimator defined by replacing $\mathrm{KT}_k(x_1^n)$ in the definition of $\hat{k}_{\mathrm{KT}}$ with the normalized maximum likelihood

$$\mathrm{NML}_k(x_1^n)=P_{\mathrm{ML}(k)}(x_1^n)\Big/\sum_{y_1^n\in A^n}P_{\mathrm{ML}(k)}(y_1^n). \tag{4.2}$$

The strong inconsistency property, (4.1), is a consequence of the following two propositions.

Proposition 5
For $k_n = \alpha \log n$, with a sufficiently large constant α,

$$-\log p_{k_n} - \log KT_{k_n}(x_1^n) + \log KT_0(x_1^n) \to -\infty, \quad a.s.$$

Proposition 6
For every fixed $k > 0$,

$$-\log KT_k(x_1^n) + \log KT_0(x_1^n) \to \infty, \quad a.s.$$

Indeed, Proposition 5 implies that $\hat{k}_{\mathrm{KT}}(x_1^n) \neq 0$, and Proposition 6 implies that $\hat{k}_{\mathrm{KT}}(x_1^n) \neq k$, for any fixed $k > 0$, both eventually almost surely.

Proof of Proposition 5. Note first that if no k-block occurs in x_1^{n-1} more than once then $\mathrm{KT}_k(x_1^n) = |A|^{-n}$. This follows from the representation (2.13) of $\mathrm{KT}_k(x_1^n)$ since the assumption on x_1^{n-1} implies that exactly $n-k$ different k-blocks a_1^k occur in x_1^{n-1}; to each of these there is one $a_{k+1} \in A$ with $a_1^{k+1} \in x_1^n$, so that $N(a_1^k|x_1^{n-1}) = N(a_1^{k+1}|x_1^n) = 1$.

The probability that some k-block occurs in x_1^{n-1} more than once is less than $n^2|A|^{-k}$. Indeed, for any ℓ, m with $1 \leq \ell < m \leq n-k+1$, the probability of $x_\ell^{\ell+k-1} = x_m^{m+k-1}$ is $|A|^{-k}$ because, arbitrarily fixing $x_1^{m-1} \in A^{m-1}$, for exactly one of the $|A|^k$ equiprobable choices of $x_m^{m+k-1} \in A^k$ will $x_\ell^{\ell+k-1} = x_m^{m+k-1}$ hold. In particular, for $k_n = \alpha \log n$ with $\alpha = 4/\log|A|$, the probability that some k_n-block occurs in x_1^{n-1} more than once is less than n^{-2}. This and the previous observation gives that

$$-\log \mathrm{KT}_{k_n}(x_1^n) = n \log |A|, \text{ eventually a. s.} \tag{4.3}$$

Next we use the fact that for any fixed $k \geq 0$ there is a constant K such that,

$$|\log P_{\mathrm{ML}(k)}(x_1^n) + n \log |A|| < K \log\log n, \quad \text{eventually a. s.,} \tag{4.4}$$

see Proposition A.2 in the Appendix, where now $\log Q(x_1^n) = -n \log |A|$. Since

$$\left| \log \mathrm{KT}_0(x_1^n) - \log P_{\mathrm{ML}(0)}(x_1^n) + \frac{|A|-1}{2} \log n \right| \leq C,$$

by the bound (2.14) in Lemma 3, it follows that $\log \mathrm{KT}_0(x_1^n)$ differs from $-n \log |A| - \frac{|A|-1}{2} \log n$ by less than (constant)$\log\log n$, eventually almost surely. This, together with (4.3), proves Proposition 5, because the hypothesis on $\{p_k\}$ implies that $\log p_{k_n} = o(\log n)$. $\square$

Proof of Proposition 6. For fixed $k > 0$, we have eventually almost surely

$$\frac{n}{2}|A|^{-k} < N(a_1^k|x_1^{n-1}) < n \quad \text{for all} \quad a_1^k \in A^k,$$

so that, by (2.16), $-\log \mathrm{KT}_k(x_1^n)$ differs from $-P_{\mathrm{ML}(k)}(x_1^n) + \frac{|A|^k(|A|-1)}{2}\log n$ by less than a constant (depending on k). Hence, using (4.4), $-\log \mathrm{KT}_k(x_1^n)$ differs from $n\log|A| + \frac{|A|^k(|A|-1)}{2}\log n$ by less than (constant)log log n, eventually almost surely. Combining this with the previous result on $\log \mathrm{KT}_0(x_1^n)$ gives Proposition 6. $\square$

Finally, to check the analogues of Propositions 5 and 6 for $\mathrm{NML}_k(x_1^n)$, we need a result from the theory of universal coding, see [16], namely, for fixed $k \geq 0$ and $n \to \infty$,

$$\log \sum_{y_1^n \in A^n} P_{\mathrm{ML}(k)}(y_1^n) \sim \frac{|A|^k(|A|-1)}{2}\log n.$$

This and (4.4) immediately give the analogue of Proposition 6 for the normalized maximum likelihood $\mathrm{NML}_k(x_1^n)$ defined by (4.2). To get the analogue of Proposition 5, we additionally need an analogue of (4.3), namely with k_n as there,

$$-\log \mathrm{NML}_{k_n}(x_1^n) \leq n\log|A|, \text{ eventually a. s.}$$

The latter follows because if no k-block occurs in x_1^{n-1} more than once then clearly $P_{\mathrm{ML}(k)}(x_1^n) = 1$, and consequently

$$\mathrm{NML}_k(x_1^n) = 1 \Big/ \sum_{y_1^n \in A^n} P_{\mathrm{ML}(k)}(y_1^n) \ \geq |A|^{-n}.$$

5 Appendix

Here we prove, for completeness, that for a stationary process $Q \in \mathcal{M}$ of order $k_0 < k^*$,

$$k_{\mathrm{BIC}}(x_1^n) \notin [0, k_0) \cup (k_0, k^*], \quad \text{eventually a.s.}$$

Clearly, this is a consequence of the following two propositions.

Proposition A.1
In the case $k < k_0$, there is a positive constant C such that

$$-\log P_{\mathrm{ML}(k)}(x_1^n) \geq -\log P_{\mathrm{ML}(k_0)}(x_1^n) + Cn, \ \textit{eventually a. s.}$$

Proposition A.2
For any fixed $k > k_0$, there is a positive constant C such that

$$-\log P_{\mathrm{ML}(k_0)}(x_1^n) \leq -\log P_{\mathrm{ML}(k)}(x_1^n) + C\log\log n, \ \textit{eventually a. s.}$$

Proof of Proposition A.1. Recall, see (2.2), that

$$\log P_{\mathrm{ML}(k)}(x_1^n) = \sum_{a_1^{k+1}\in x_1^n} N(a_1^{k+1}|x_1^n)\log\frac{N(a_1^{k+1}|x_1^n)}{N(a_1^k|x_1^{n-1})}.$$

The ergodic theorem implies that $N(a_1^{k+1}|x_1^n)/(n-k)$ converges almost surely to $Q(a_1^{k+1})$, for each $a_1^{k+1}\in\mathcal{S}_{k+1}$, as $n\to\infty$, and hence

$$\lim_{n\to\infty}\left(-\frac{1}{n}\log P_{\mathrm{ML}(k)}(x_1^n)\right) = -\sum_{a_1^{k+1}\in\mathcal{S}_{k+1}} Q(a_1^{k+1})\log\frac{Q(a_1^{k+1})}{Q(a_1^k)},\quad \text{a. s.,} \tag{A.1}$$

the limit being the conditional entropy H_k of X_k, given X_1^{k-1}. (Here $Q(a_1^k)$ is replaced by 1 if $k=0$.) It is well known, see [15, Theorem I.6.11], that H_k is strictly greater than H_{k_0} if $k<k_0$. Hence (A.1) implies the proposition. □

Proof of Proposition A.2. Fix $k>k_0$. Since Q is a stationary process in $\mathcal{M}_k$, the product formula (2.1) yields

$$\log Q(x_1^n) = \log Q(x_1^k) + \sum_{a_1^{k+1}\in x_1^n} N(a_1^{k+1}|x_1^n)\log Q(a_{k+1}|a_1^k)$$

whenever $Q(x_1^n)>0$. This and the expression (2.2) for $\log P_{\mathrm{ML}(k)}(x_1^n)$, yield

$$\begin{aligned}&\log P_{\mathrm{ML}(k)}(x_1^n) - \log Q(x_1^n) = \\ &\quad -\log Q(x_1^k) - \sum_{a_1^{k+1}\in x_1^n} N(a_1^{k+1}|x_1^n)\log\frac{Q(a_{k+1}|a_1^k)}{N(a_1^{k+1}|x_1^n)/N(a_1^k|x_1^{n-1})}\end{aligned} \tag{A.2}$$

(with an obvious modification if $k=k_0=0$).

Recall from Proposition 3 in Section 3 that the ratio

$$\frac{\hat{P}_n(a_1^k)}{Q(a_1^k)} = \frac{N(a_1^k|x_1^n)}{(n-k+1)Q(a_1^k)}$$

differs from 1 by less than (constant) $\sqrt{\frac{\log\log n}{n}}$, eventually almost surely. Applying this with k replaced by $k+1$, and also with n replaced by $n-1$, it follows, since $Q(a_{k+1}|a_1^k) = Q(a_1^{k+1})/Q(a_1^k)$, that the quantity

$$\varepsilon(a_1^{k+1}|x_1^n) = \frac{Q(a_{k+1}|a_1^k)}{N(a_1^{k+1}|x_1^n)/N(a_1^k|x_1^{n-1})} - 1$$

is bounded in absolute value by (constant) $\sqrt{\frac{\log\log n}{n}}$, eventually almost surely. Hence, using that $\log(1+\varepsilon)=\varepsilon+O(\varepsilon^2)$, the summation in (A.2) becomes

$$\sum_{a_1^{k+1}\in x_1^n} N(a_1^{k+1}|x_1^n)\varepsilon(a_1^{k+1}|x_1^n) + \eta_n,$$

where $|\eta_n| \leq$ (constant) $\log\log n$, eventually almost surely. Eventually almost surely, however, the last summation is over all $a_1^{k+1} \in \mathcal{S}_{k+1}$, so that it takes the form

$$\sum_{a_1^{k+1} \in \mathcal{S}_{k+1}} [Q(a_{k+1}|a_1^k)N(a_1^k|x_1^{n-1}) - N(a_1^{k+1}|x_1^n)].$$

This sum is equal to 0, since summing for a_{k+1} with $a_1^k \in \mathcal{S}_k$ fixed gives zero. This completes the proof of Proposition A.2.

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